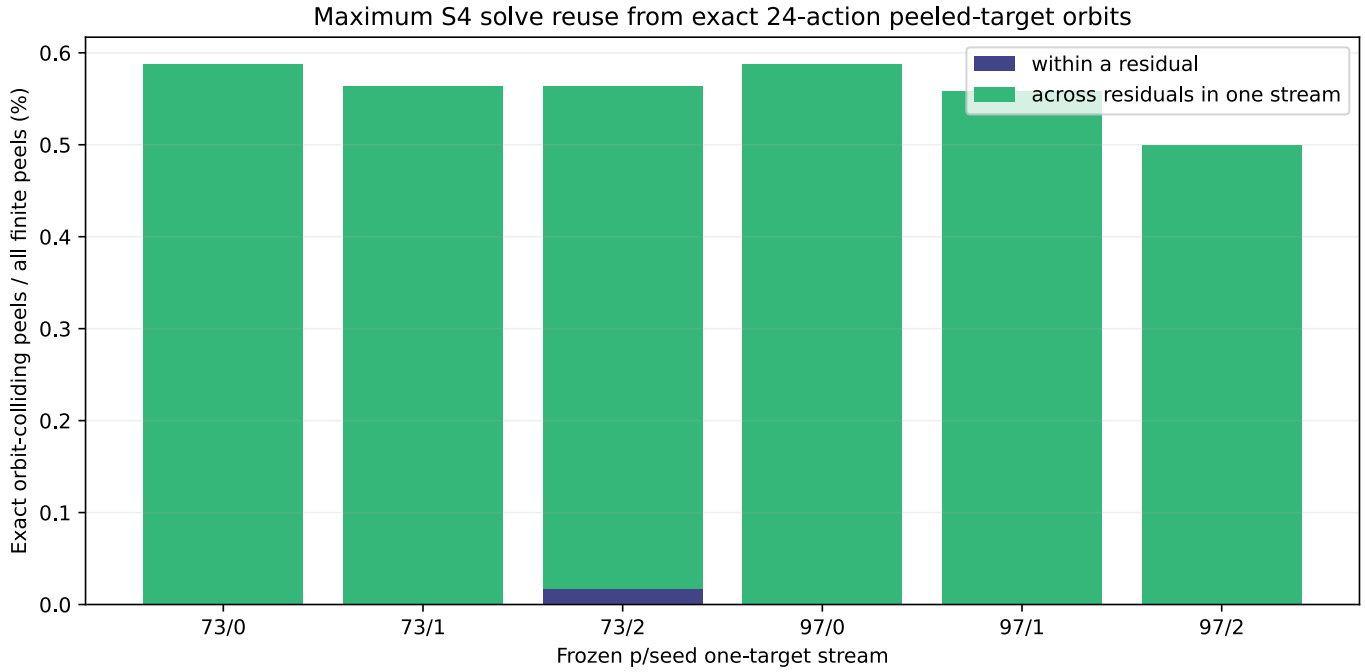
Can twist symmetry reuse peeled S4 systems?

23 October 2026. Six complete frozen p=73/97 one-target streams. Protocol commit 8801c22aaa; source and raw record in the study directory. For each residual and signed L2 factor, canonicalize R−T under the verified 24-element Frobenius/cubic action. Equal canonical targets have symmetry-equivalent S4 solution sets in principle. This counts possible calls eliminated before charging the cache. For a nonzero residual R, corresponding factors T and λT in one L2 orbit cannot give λ-equivalent peeled targets: R−λT=λ(R−T) would imply (1−λ)R=0, impossible for λ≠1 in the prime-order group. All 1,856 tested residuals are nonzero.

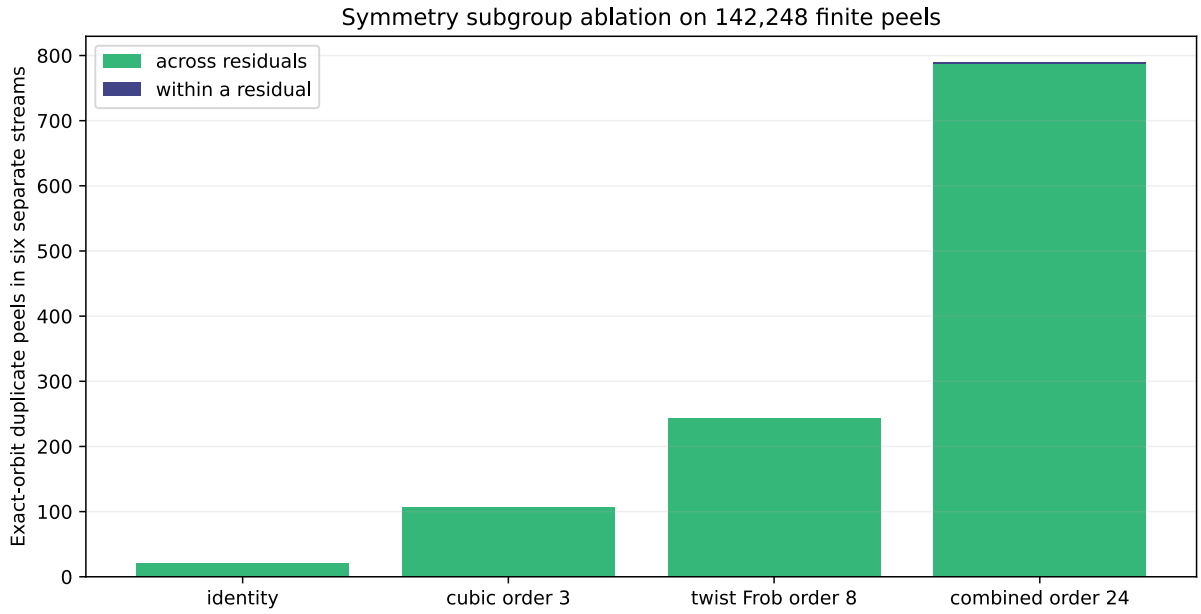


p/seed	peels	local dup	cross dup	max cache ratio
73/0	11,232	0	66	1.005911
73/1	11,520	0	65	1.005674
73/2	11,712	2	64	1.005667
97/0	40,176	0	236	1.005909
97/1	32,976	0	184	1.005611
97/2	34,632	0	173	1.005020

Across separate streams: 142,248 finite peels, two within-residual and 788 additional within-stream exact-orbit collisions. The ideal maximum call-count ratio is 1.005585. Group-action closure, all-image canonical equality and a perturbed-eigenvalue rejection were checked. The physical logs came from the prior independently audited BSGS records.

Subgroup ablation

Follow-up protocol commit 433df0f52d. Identity, cubic order 3, transported Frobenius order 8 and combined order 24 are compared on the same six streams. The full-24 counts reproduce the first record.



action	order	local	cross query	ideal ratio
identity	1	0	20	1.000141
cubic	3	0	107	1.000753
frob	8	0	243	1.001711
full24	24	2	788	1.005585

Combined action identifies 770 more targets than identity-only caching among 142,248 finite peels. It identifies 683 more than cubic alone. These counts exclude cache work. This is an exact orbit count, not a measured polynomial cache or whole-method speed result. Other factor bases and curves remain open.